AI Exposure and U.S. Employment After 2022

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A note before you read: I am not an economist, and this has not been peer reviewed. I kept hearing confident claims about AI and jobs, got curious, and spent some of my free time seeing what public data and what I know about causal inference could actually show. I tried to be careful and upfront about where this falls short. If you find something I got wrong, I would like to hear it.

This is the technical version. A shorter version for general readers, AI or Interest Rates?, has the same results and is in the same repository.

Abstract

This side project asks whether employment in work more exposed to generative AI fell, relative to less exposed work, after the release of ChatGPT. The question is motivated by the decline in U.S. technology hiring, but the paper does not measure technology hiring directly: its outcome is total employment across occupations and private industries, and exposure captures what a language model could do rather than observed use. Because ChatGPT coincided with rising interest rates, the end of a pandemic hiring boom and a tax change, I compare more and less exposed work using three public sources: the Current Population Survey (CPS), the Current Employment Statistics payroll survey (CES) and the Quarterly Workforce Indicators (QWI). Within sectors, the pre-registered design estimates an average 2024–26 change, relative to 2022, of +0.3% between the top and bottom exposure quartiles (95% CI −5.6% to +6.1%). Across industries, it estimates −5.0% (95% CI −9.0% to −0.9%), but that design fails its pre-trend test, so the pre-registered rules do not interpret it; weighting industries by employment gives −5.5%, although the same model also shows a decline in the dot-com bust. In more exposed work, young workers fell 6.3% relative to older workers ($p = 0.18$), but that estimate ranges from about −1% to −11% across exploratory checks. The pre-registered rules return "can't tell yet," and because Test A's pre-trend test sits near its threshold, the estimates are a better guide than the label (Section 6.6). The study does not establish how many jobs AI has added or removed, or that AI has no effect on employment.

Keywords: generative AI, labor markets, difference-in-differences, synthetic control, event study, pre-registration, causal inference

1. Introduction

There has been considerable debate about AI and employment. Some expect AI to make many jobs obsolete; others expect it to resemble the steam engine and ultimately create more work than it displaces. The honest answer is that nobody knows yet, myself included.

The common argument, that software jobs have fallen since ChatGPT and AI must therefore be responsible, compares one group with itself over time. That is a confounded comparison, because it credits AI with everything else that changed during the same period. Several alternatives are plausible: interest rates rose sharply, firms that over-hired during the pandemic began to retrench, and a tax change raised the after-tax cost of developers. The important detail is that these shocks did not all arrive at the same time, and that variation in timing is what makes the question testable.

Table 1: Timeline of the principal shocks.

Date	Event
Tax years from 2022	Section 174 requires R&D costs, including software development costs, to be amortized over five years rather than deducted immediately
March 2022	The Federal Reserve begins raising rates from near zero, reaching a 5.25–5.50% target in July 2023
Through 2022	Indeed reports software postings already falling "as tech companies rein in hiring plans" [13]
November 30, 2022	ChatGPT is released
Tax years from 2025	A July 2025 law restores immediate expensing of domestic R&D

I ask whether exposed work fell across industries, within sectors and among young workers, and whether interest rates, the boom or the tax change can account for what appears. Three features may make the exercise useful: it tests the leading early-career finding on public data anyone can reproduce, it uses the dot-com bust as a placebo, and it measures what each design can detect, so that a null result is not read as more than it is.

1.1 What This Paper Does and Does Not Measure

The scope of this study is narrower than the debate that motivates it. The main results concern total employment across all occupations and private industries, not the technology sector, and they are relative comparisons. Table 2 states these boundaries directly, and every result below should be read within them.

Table 2: Scope of the study.

Dimension	What this paper measures	What it does not measure
Outcome	Employment counts, meaning how many people hold jobs; hires in the QWI analysis only	Technology hiring, job postings, layoffs, wages, hours or job quality
Population	Total employment at all exposure levels: occupations across sectors, ages 18–65 (CPS), and private industries (CES)	The tech sector or software developers on their own, apart from one inconclusive estimate for computer systems design
Treatment	Potential exposure: tasks a language model could speed up	Whether employers or workers actually use AI
Comparison	How more exposed work changed relative to less exposed work	How many jobs AI added or removed in total, or effects that reach all work equally
Causal claim	Whether patterns are consistent with an AI effect, under stated assumptions	Proof that AI did or did not cause a change, or how much rates, the boom and taxes each contributed
Time and place	The United States, with 2024–26 compared with 2022, through September 2026	Other countries or long-run effects

In practical terms, a null result here means that the average change in the employment share of jobs AI could do more of stayed within the reported bounds. It does not mean that AI left tech hiring, or employment overall, unchanged.

2. Related Work

Brynjolfsson, Chandar and Chen [1] use payroll records through June 2026 from ADP, a large payroll processor, and find no evidence of widespread, economy-wide displacement. The exception is workers aged 22 to 25 in AI-exposed occupations, whose employment stands 19% below where it would be had it

kept pace with less exposed peers, up from 15% in July 2025 data on the same measure. Earlier versions of the paper headlined firm-adjusted regression estimates of 13% and 16%, which are a different measure. Experienced workers show no comparable gap, and the divergence operates mainly through reduced hiring rather than increased separations.

The authors treat these patterns as descriptive rather than causal. The divergence persists when controlling for interest-rate exposure and remote work, attenuates when controlling for education, shows some trends that predate generative AI, and is more pronounced in their ADP sample than in national survey benchmarks such as the CPS. Since payroll microdata is proprietary, I ask how far the same question can go with public data.

An earlier note by the same authors [2] finds that occupational rate sensitivity, as estimated by Zens et al. [10], is negatively correlated with AI exposure, and that the exposed decline appears at both high and low rate sensitivity. With firm-by-time fixed effects, however, the relative decline becomes significant only after 2024, and they describe the earlier declines as "likely (at least partly) due to some combination of other factors, not just AI." Their later version notes that those results were sensitive to changes in their data pipeline [1]. This motivates Test B, whose sector-by-year fixed effects are a much coarser public-data analogue.

Eloundou et al. [3] provide the exposure measure used here and in [1], which keeps the results comparable. Yin, Vu and Persico [9] show that LLM-based exposure scores can agree as little as 57% across models, with difference-in-differences coefficients varying 2.4-fold depending on the model used for scoring. I therefore average GPT-4 and human ratings rather than relying on a single rater, and I treat measurement error in the treatment as a known limitation.

Others have asked related questions with public data. Gimbel et al. [18] compare the pace of change in the CPS occupational mix since ChatGPT with earlier technology shifts, and find that measures of exposure, automation and augmentation show no sign of being related to changes in employment or unemployment. Atkinson and Yamco [19] use the CPS and the exposure measure in [3], and find that the employment share of workers aged 20 to 24 in the most exposed occupations fell after ChatGPT, mainly because fewer new labor-market entrants found jobs there; they caution that a factor correlated with exposure, such as education, could be responsible. Frank et al. [20], using unemployment-insurance records and LinkedIn profiles, find that unemployment risk in exposed occupations began rising in early 2022, before ChatGPT's release, and argue that the release is a flawed natural experiment. This study differs mainly in its design: pre-registered comparisons at the industry as well as the occupation level, explicit controls for interest-rate sensitivity and the pandemic hiring boom, and a dot-com placebo.

Methodologically, this study draws on synthetic control [5], synthetic difference-in-differences [4], and sensitivity analysis for violations of parallel trends [8].

3. Data

The analysis requires employment by job type, a measure of AI exposure, and a means of accounting for interest-rate sensitivity. No single source provides all three, so I combine three employment datasets selected for their complementary strengths with several supporting sources. The Current Population Survey (CPS) is the monthly household survey the Census Bureau conducts for the Bureau of Labor Statistics (BLS), and the source of the unemployment rate; I use the harmonized version distributed by IPUMS [11]. The Current Employment Statistics (CES) survey is the BLS monthly survey of employer

payrolls, which I download from Federal Reserve Economic Data (FRED), the St. Louis Fed's public database. The Quarterly Workforce Indicators (QWI) are compiled by the Census Bureau from state unemployment-insurance records and count actual hires. Industries are coded with the North American Industry Classification System (NAICS), and occupational tasks come from O*NET, the Department of Labor's occupational database.

Table 3: Data sources.

Source	Variables used	Rationale	Limitations
IPUMS CPS [11], Basic Monthly, 1990–Sep 2026	Employment by age, harmonized occupation (OCC2010) and industry (IND1990)	The only free monthly source with occupation, industry and age together, extending through the dot-com period	Small samples per occupation; occupation coding changes, including 2003 and 2020 [14, 15]; October 2025 not collected; microdata cannot be redistributed
BLS CES via FRED, 1990–Sep 2026	Monthly payroll employment for 99 detailed private industries	Large employer samples and consistent NAICS definitions through the dot-com bust	Industry rather than occupation; revised annually; Test A's industries cover about 86% of private payrolls
Census QWI	Hires by age band, across 27 NAICS industries and 4 Census regions	Administrative records of actual hires, the margin on which [1] finds the effect	Lags by about three quarters; residual seasonality, which I remove
Eloundou et al. [3]	Beta exposure scores, GPT-4 and human ratings	The standard LLM exposure measure, used in [1]	Measures potential rather than actual use; raters disagree
FRED; Bauer–Swanson surprises [6]	Interest rates, the consumer price index (CPI), and orthogonalized monetary policy surprises	Surprises isolate rate changes from the economic conditions that prompt them	The series ends in December 2023; it is used only to estimate 1990–2019 sensitivity, which is then interacted with year indicators
Census and O*NET crosswalks	Mappings between occupation and industry code systems	Required to attach exposure scores to CPS occupations and industries	1.8% of employment unmatched

The pairing of CPS and QWI is deliberate. CPS observes occupation and age but contains only a few hundred software developers per month. QWI offers large samples of actual hires by age but observes only industry. Agreement between the two would be considerably stronger evidence than either provides alone.

Exposure. The treatment variable is a continuous dose. For each occupation, I average the z-scored GPT-4 and human beta ratings, then re-standardize so that one unit equals one standard deviation. Beta counts tasks an LLM can speed up directly, plus half of those it can speed up with additional software. Industry exposure is the employment-weighted average across each industry's 2015–19 occupation mix, as pre-registered, so the mix predates ChatGPT. CPS industry codes changed in 2020, so I apply the Census list of 2012-to-2017 code changes to match the 2017 crosswalk. Two CES series are aggregates with no NAICS code of their own (Motor Vehicles and Parts; Beverage, Tobacco and Leather), so I map them to the NAICS industries they combine, and Census "not specified" rows are used only for an industry with no specific match. An earlier version of this analysis used a 2020–25 mix, which departed from the pre-registration and was partly post-treatment, and it also lacked those two matching corrections; Section 5 reports what correcting each changed. Occupation scores still use 2020–25 weights to map Census codes to OCC2010, because the two codes coexist only from 2020; using 2020–22 weights moves no occupation's GPT-4 score by more than 0.01. The measure captures what AI could do rather than what it is doing, so every finding should be read as applying to jobs where AI could do more of the work.

Rate sensitivity and cyclicality. For each industry and occupation group, I estimate sensitivity to monetary policy surprises and to the business cycle on 1990–2019 data only, before generative AI existed, so AI cannot contaminate either control. Estimates are shrunk toward zero with empirical Bayes to reduce noise. The estimation window overlaps the event-study pre-periods, and the dot-com placebo applies estimates that include later years. For occupation groups, shrinkage reduces rate sensitivity to a constant, leaving no usable variation, which is why Test B controls only for cyclicality.

I also downloaded the Indeed Hiring Lab postings index, which is available under CC BY 4.0. It is not used in any reported result, because it begins in 2020 and measures postings rather than employment or hires.

Checks against published data. My CPS extract covers ages 18 to 65. At ages 25–54, its weighted employment stays within 0.7% of the published BLS series in every month since 2010. For seven of nine occupations and occupation groups checked, the 2019–25 change is within 3 percentage points of BLS annual tables. The exceptions are software developers (+29% versus +24%) and management occupations (+17% versus +10%), which may reflect the harmonized occupation coding discussed in Section 8.

4. Methodology

The fundamental problem is that 2022 cannot be rerun without ChatGPT, so every design must construct a credible counterfactual. I use several designs because their weaknesses differ, an approach known as triangulation. A confounder capable of fooling one design is unlikely to fool them all in the same direction.

4.1 Round 1: Synthetic Control

Synthetic control [5] builds a weighted combination of untreated units that tracks a treated unit before the intervention and treats the later gap as the effect. A first specification, computer occupations in the CPS against 36 low-exposure groups, failed its hold-out test with an error of 25.2%, about five times its fit error, and I kept it as a documented negative result. I moved to the computer systems design industry in CES and fixed a selection rule before comparing variants: the lowest 2018–19 hold-out error among variants with at least four effective donors. The chosen variant uses 67 donors with a ridge penalty of 0.3, a regularization not part of [5], and has a hold-out error of 0.6%. Inference comes from placebo runs on donor industries.

4.2 Round 1: Synthetic Difference-in-Differences

A single treated unit can reliably detect only effects of about 16 to 20%, so I pooled units with synthetic difference-in-differences (SDiD) [4]. In CES, treated industries are in the top quartile of both exposure ratings, and controls are below the median on both. In QWI, which has fewer industries, treated industries are above the median on both ratings and controls are below it on both. The QWI outcome is log hires at ages 22–24 minus log hires at ages 35–44. SDiD combines unit weights, which align the controls' pre-treatment path with the treated group's, and time weights, which emphasize the pre-treatment periods most predictive of the post period:

$$\hat{\tau} = \left(\bar{Y}_{\mathrm{tr}}^{\mathrm{post}} - \sum_{t \le T_0} \hat{\lambda}_t \, \bar{Y}_{\mathrm{tr},\,t}\right) - \sum_{j} \hat{\omega}_j \left(\bar{Y}_{j}^{\mathrm{post}} - \sum_{t \le T_0} \hat{\lambda}_t \, Y_{jt}\right) \tag{1}$$

Here, the omega terms are the unit weights over control units j, the lambda terms are the time weights over pre-treatment periods, and T0 is the last pre-treatment period. The estimator is implemented directly in the repository rather than taken from a package, and it is tested on simulated data with known effects. Standard errors use the placebo procedure of [4], with a jackknife by industry for QWI, where treated and control units are too close in number for the placebo procedure.

Two decisions shaped these estimates. First, I excluded 2020 Q1 through 2021 Q2, because COVID affected in-person, low-exposure industries for reasons unrelated to AI or rates. Second, the data-driven time weights placed 81% of their mass on 2021 Q4, the peak of the pandemic boom, which turned the estimate negative by measuring change from the boom's peak. I therefore adopted a 2019 baseline as the primary specification and report the late-2021 version alongside it. Both choices were made after seeing data, and the baseline change moved the estimate from −3.9% to +0.7%, toward a null, so readers should weigh it accordingly. That risk is what motivated pre-registering Round 2.

4.3 Round 2, Test A: Dose-Response Across Industries

Splitting units into high and low exposure discards information, since an industry in the middle of the distribution is simply dropped. Test A instead uses exposure as a continuous treatment across the 94 industries observed in every quarter. The panel has 41 quarters: the 47 from 2015 Q1 to 2026 Q3, less 2020 Q1 through 2021 Q2:

$$\log E_{it} = \alpha_i + \delta_t + \sum_{k \neq 2019} \theta_k X_i \mathbf{1}[\text{year}(t) = k] + \sum_{k \neq 2019} \phi_k R_i \mathbf{1}[\text{year}(t) = k] + \sum_{k \neq 2019} \psi_k C_i \mathbf{1}[\text{year}(t) = k] + \varepsilon_{it} \tag{2}$$

E is employment in industry i and quarter t. The terms alpha and delta are industry and quarter fixed effects, X is exposure, and R and C are rate sensitivity and cyclicality, whose year-specific coefficients allow rate-sensitive and cyclical industries to follow their own paths. Each theta measures the employment gap, relative to 2019, between industries one standard deviation apart in exposure. Because delta absorbs aggregate shocks, the design is relative by construction: it can show that more exposed industries did worse, but not how many jobs AI affected in total.

The pre-registered estimand compares the adoption period with 2022, a year that had rate hikes but effectively no generative AI adoption:

$$\Delta_{\text{AI}} = \tfrac{1}{3} \sum_{k=2024}^{2026} \theta_k - \theta_{2022} \tag{3}$$

I scale this to a top-versus-bottom quartile comparison using the observed gap of 2.52 standard deviations. Identification requires that, absent AI, more and less exposed industries would have kept the same relative path after 2022, conditional on R and C. The main threat is another shock that hit exposed industries from 2023 onward. A further caveat is that exposed industries stood at their post-2019 high in 2022, so this contrast can capture the continued unwinding of the boom.

4.4 Round 2, Test B: Occupations Within Industries

Industry totals cannot detect an accounting firm replacing junior bookkeepers while hiring more sales staff. Test B therefore compares occupations inside the same sector, across 576 cells formed from 108 occupation groups and 14 sectors, annually from 2011, dropping 2020 and the first half of 2021:

$$\log E_{ojt} = \alpha_{oj} + \delta_{jt} + \sum_{k \neq 2019} \theta_k X_o \mathbf{1}[t = k] + \sum_{k \neq 2019} \psi_k C_o \mathbf{1}[t = k] + \varepsilon_{ojt} \tag{4}$$

The key term is the sector-by-year fixed effect, delta, which absorbs anything that hit all occupations in sector j equally in year t. It does not absorb shocks that differ by occupation within a sector, such as rate sensitivity [10], Section 174 or an uneven pandemic boom. The specification controls for occupational cyclicality but not rate sensitivity; adding unshrunk rate-sensitivity estimates leaves the result at +0.2%. Employment in a cell is the sum of the CPS final person weight (WTFINL) across employed respondents, divided by the months surveyed that year, and the outcome is its logarithm. Cells are weighted by 2015–19 employment and standard errors are clustered by occupation group, with no separate adjustment for the CPS sample design. Cells must have at least 30 respondent-months in every year, which retains 97% of 2015–19 employment; because that filter includes post-ChatGPT years, Sections 6.2 and 6.3 also apply it only through 2019. The top-versus-bottom quartile gap in this sample is 2.55 standard deviations. Beginning in 2011 avoids the 2003 occupation coding break.

A secondary age version compares young and older workers (ages 22–27 and 35–65), where a indexes the age group:

$$\log E_{ojat} = \alpha_{oja} + \delta_{jat} + \sum_{k \neq 2019} \theta_k X_o \mathbf{1}[t = k] + \sum_{k \neq 2019} \lambda_k X_o \, \mathrm{Young}_a \mathbf{1}[t = k] + \varepsilon_{ojat} \tag{5}$$

The terms alpha and delta are now cell-by-age and sector-by-age-by-year fixed effects, and each lambda measures how much the exposure gradient differs for young workers. The result is a triple difference across exposure, age and time, the closest analogue to [1] in these data, although its 22–27 age band is broader than their 22–25. It uses 651 cell-by-age units (167 young and 484 older) and omits the cyclicality control, my reading of a plan that did not settle the point; adding that control gives −6.5%. Its quartile gap is 2.53 standard deviations. This design cannot by itself separate AI from other causes of a hiring slowdown. Young workers enter through hiring and age out of the band each year, so any shock that slows hiring in an occupation lowers this estimate.

4.5 Placebos and Injection Tests

The dot-com bust serves as a placebo: the Test A and SDiD designs applied to a technology downturn with no generative AI, where 2001 stands in for 2022 and 2002–03 for the adoption period. Test B cannot be run on this period because of the 2003 coding change. The Test A version also fails its own pre-trend test ($F = 8.41$, $p = 0.0004$) on only two pre-period years, so I treat the comparison as suggestive. It is a calibration rather than a perfect twin, since the Federal Reserve was cutting rates in 2001 and raising them in 2022. I also run fake-date placebos, assigning a false treatment date in 2016 or 2017 using only pre-2020 data.

Injection tests measure statistical power empirically, using the real data's noise. I plant a known effect and rerun the exact pipeline. Round 1 plants a sudden step in relabeled control units, 100 times per size. Round 2 plants a linear ramp beginning in 2023, on a shuffled exposure score, 200 times per size. A ramp is the more demanding test, because adoption is gradual. The pre-registered rule states that a design whose 80%-power detectable effect exceeds 10% can report only an uninformative null, which prevents a weak test from being presented as evidence of no effect.

5. Inference and Decision Rules

Round 1 involved choices made after seeing data, the problem Gelman and Loken call the garden of forking paths [7]. Before Round 2, I therefore froze the questions, specifications, inference and decision rules in a pre-registration and recorded the file's SHA-256 hash, which docs/PREREGISTRATION_STATUS.md lists. It is not a pristine pre-registration: CES, QWI and CPS data had been examined in Round 1, and the age version uses the same CPS data in which Round 1 had shown a young-worker decline.

Standard errors. Errors are clustered by unit, because a unit's observations over time are not independent. The clusters are the 94 industries in Test A and the 108 occupation groups in Test B.

Robust p-values. Headline estimates also report a wild cluster bootstrap (Rademacher weights, 999 draws), since cluster-robust formulas can be unreliable with few clusters, and randomization inference (exposure shuffled across units 1,000 times), which avoids assumptions about the error distribution but requires exposure labels to be exchangeable under the null. Exposure was not randomly assigned, so that assumption is imperfect. In Test B, shuffling only within groups of similar education changes little ($p = 0.89$ versus 0.91). The three procedures address different statistical concerns: agreement suggests the conclusion is not an artifact of one of them, but none removes the identifying assumptions.

Multiple testing. Tests A and B are the two primary hypotheses, and their p-values receive a Holm correction [12]. Everything else is labeled secondary and cannot, on its own, support a positive verdict.

Bayesian probability. I report the probability that the true effect is below zero under a flat prior and, for headline tests, under a skeptical prior centered on zero with a standard deviation of 2.5%. The flat-prior figure restates the p-value on an easier scale rather than adding evidence, and it speaks to the sign of the change, not its cause. I added this after seeing Test A, and I record it as a deviation from the plan.

Pre-trend test. Before the event, exposure should not predict diverging employment paths. I test whether the pre-period coefficients are jointly zero, using 2015–18 for Test A and 2012–18 for Test B and its age version, with a pass threshold of $p > 0.10$. Such omnibus tests have limited power against a steady drift, so I also report a linear-slope test for Test B and the age version.

Table 4: Verdicts, fixed before Round 2 was run.

Verdict	Requirements
1. Evidence AI is reducing employment in exposed work	A primary estimate is negative and significant after Holm correction; the 2022 coefficient is not already half the size of the 2024+ average; the result differs from the dot-com bust or passes the fake-date placebo; at least 70% of the specification curve shares its sign; and at least two secondary fingerprints agree
2. Evidence AI is increasing employment in exposed work	The same requirements with the sign reversed

Verdict	Requirements
3. No effect larger than X so far	Both primaries pass their checks and neither is significant, where X is the larger confidence bound in absolute value
4. Can't tell yet	Any other outcome, including a significant result that fails the timing or history rule

The first run of Test A used a 2020–25 occupation mix rather than the pre-registered 2015–19 mix, and it returned a pre-trend p-value of 0.078 against a threshold of 0.10, which I reported as a failure. A later review identified the departure, and the registered mix returned 0.107, a narrow pass. An audit then found two errors in how industries were matched to occupation data (Section 3). With both corrected, the registered mix gives −5.0% and a pre-trend p-value of 0.073, a failure. Across the six versions of the industry score I ran, the pre-trend p-value ranges from 0.067 to 0.107 and only one passes, while the estimate stays between −3.7% and −5.0%. I report the corrected, registered version, and every correction came after earlier results were known. The verdict label therefore rests on a test near its threshold, and the estimates are more stable than the label.

6. Results

Estimates compare the top and bottom exposure quartiles unless noted, using data through September 2026 (2025 for QWI). Unless noted, the outcome is total employment across occupations and industries rather than technology hiring, and each estimate is relative to less exposed work (Section 1.1). Percentages are approximations from differences in logs.

Table 5: Principal estimates. Event-study rows compare the top and bottom exposure quartiles; the synthetic DiD row compares industries above and below the median on both exposure ratings.

Comparison	Design and data	Estimate	95% CI	Key diagnostic	Interpretation
Occupations within the same sector	Test B: event study, CPS	+0.3%	−5.6% to +6.1%	Pre-trend p = 0.23 (passed)	Bounded null
All industries, by exposure	Test A: event study, CES	−5.0%	−9.0% to −0.9%	Pre-trend p = 0.073 (failed)	No verdict
Hires, ages 22–24 vs 35–44	Synthetic DiD, QWI	−9.3%	−18.3% to −0.2%	Pre-period fit error (RMS) 4.5%	Precedes ChatGPT
Employment, ages 22–27 vs 35–65, same occupation and sector	Triple difference, CPS	−6.3%	−15.5% to +2.9%	Pre-trend p = 0.18 (passed)	Inconclusive

6.1 Across Industries

SDiD across 20 high-exposure and 43 low-exposure industries estimates +0.7% (95% CI −2.9% to +4.3%) with a pre-period fit error, measured as the root mean square (RMS) of the pre-treatment gaps, of 0.6%. Exposed industries stood about 5.8% above normal by 2022 and returned to about −3.9% by 2026. Moving the baseline to late 2021 produces a significant −3.9%, which illustrates how readily the peak of a boom can produce an apparent decline. The same concern applies in part to Test A, whose estimate is measured from 2022, when exposed industries stood at their post-2019 high of +3.0% per standard deviation relative to 2019.

Test A estimates −5.0% (95% CI −9.0% to −0.9%), with p-values of 0.017 (clustered), 0.016 (wild bootstrap) and 0.010 (randomization), as shown in Figure 1. The probability of a negative effect is 99% under a flat prior and 97% under the skeptical prior. The slide sits in the more exposed half rather than

rising smoothly with exposure: relative to the lowest exposure quintile, the second through fifth quintiles are at +1.7%, +0.4%, −4.0% and −3.9%. Taken alone, this is the pattern AI would produce, beginning in 2023 rather than 2022. I do not treat it as evidence of an AI effect, but I also cannot rule one out. Four considerations apply:

1. **The pre-trend test** fails ($F = 2.22$, $p = 0.073$), so the pre-registered rule returns no verdict for Test A. Exposed industries were rising relative to others before 2019, so the violation may mean that −5.0% understates the shortfall rather than overstates it.
2. **It would survive the multiple-testing correction.** The smaller primary p-value, 0.017, is below the 0.025 that Holm requires, but a design that fails its pre-trend test is reported as failed rather than interpreted.
3. **About 40% of it runs through the boom.** Controlling for how far each industry sat above its 2015–19 trend in 2022 reduces the estimate to −2.9% (95% CI −7.1% to +1.2%), and this version passes the pre-trend test ($p = 0.28$). This control could also absorb a true AI effect, if AI is what ended the boom, so the gap between the two is best read as the portion the design cannot attribute.
4. **The dot-com bust shows a broadly similar rise and slide.** The same model yields −3.9% (95% CI −6.6% to −1.1%) with no generative AI, and the difference between the two busts is −1.1%, for which I have no confidence interval. This is suggestive only, because the dot-com model clearly fails its own pre-trend test.

An exploratory estimate that extrapolates the 2015–19 trend, the simplest case of the sensitivity analysis in [8], gives a larger shortfall of −6.1% (95% CI −9.5% to −2.7%). I added it after seeing the earlier mix's pre-trend failure and do not use it for the verdict. The trend was flattening by 2018–19, so a straight line probably overstates the shortfall, but the result shows that the industry conclusion depends on how the pre-trend is treated.

Two other checks belong beside this. A fake treatment date in 2017 gives −0.1% (95% CI −2.2% to +2.0%). Weighting industries by 2015–19 employment, a robustness check the pre-registration names but I first omitted, gives −5.5% (95% CI −9.0% to −2.1%, pre-trend $p = 0.104$), with wild bootstrap and randomization p-values below 0.01. Leaving out any single industry keeps it between −4.6% and −6.4%. This is the clearest evidence of a relative decline in the paper, and it describes the typical job rather than the typical industry. It also needs caution: the boom control reduces it to −3.1% (95% CI −6.6% to +0.3%), and the same weighted model gives −4.0% (95% CI −6.3% to −1.6%) in the dot-com bust, although that placebo fails its pre-trend test ($p = 0.027$). The unweighted estimate remains primary because it was specified in advance.

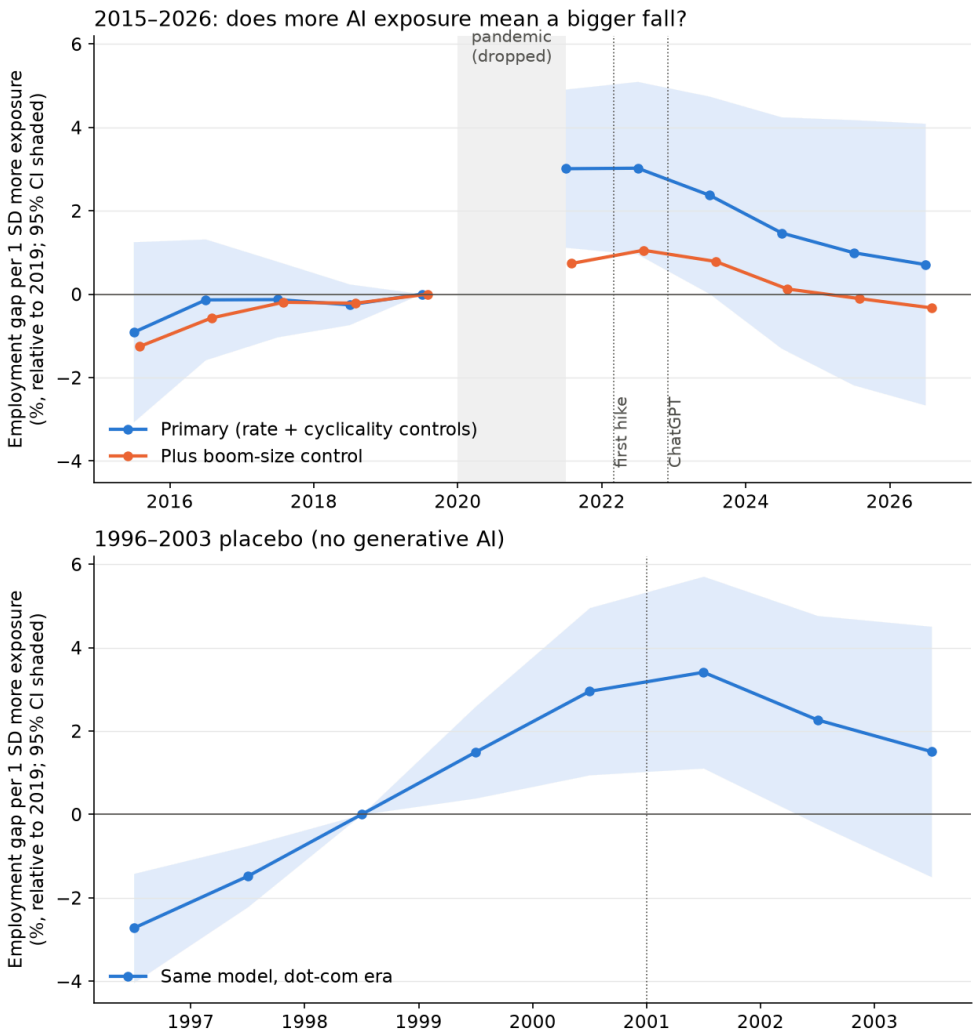


Figure 1: Test A coefficients per standard deviation of exposure, relative to 2019, with 95% confidence intervals. The top panel shows 2015–2026 and the bottom panel applies the same model to 1996–2003. Both show exposed industries rising and then sliding back. Multiplying the change between 2022 and 2024–26 by the quartile gap of 2.52 gives the headline estimate.

For a single industry, computer systems design sits 13.0% below its synthetic control in 2026 (placebo p = 0.29), with gaps of +0.2% in 2022 and −7.2% in 2024. The timing is consistent with AI, but the gap falls within the range of placebo industries, the largest donor (home health care) carries 31% of the weight, and excluding it gives −10.5%. I treat this result as inconclusive.

6.2 Within Industries

This is the most informative result for relative employment across all ages. Test B estimates +0.3% (95% CI −5.6% to +6.1%), comparing 2024–26 with 2022, as shown in Figure 2. The clustered, wild bootstrap and randomization p-values are 0.93, 0.93 and 0.91. The probability of a negative effect is 47% under a flat prior and 48% under the skeptical prior, which is effectively a coin flip. The pre-trend test passes (F = 1.36, p = 0.23).

Table 6: Test B robustness checks.

Specification	Estimate	95% CI
Primary	+0.3%	−5.6% to +6.1%
Excluding computer occupations	−0.4%	−6.6% to +5.8%
Excluding information and professional services	0.0%	−6.2% to +6.2%
Removing each cell's linear trend	−1.9%	−8.0% to +4.2%
Fake treatment date, 2016	+0.1%	−3.6% to +3.9%
Cell filter applied through 2019 only (exploratory)	−1.0%	−7.1% to +5.1%
College share by year added (exploratory)	−0.5%	−8.4% to +7.4%

A null is informative only if the design could have detected an effect. Ramp injections on a shuffled exposure score show 88–90% power against a gradual decline reaching 10% by the final period, which satisfies the pre-registered power rule as measured. However, the shuffled estimates vary by 2.2%, while the actual estimate has a standard error of 3.0%. Using the actual standard error, the 80%-power threshold is about 8% on the estimand, or a ramp reaching about 13%, so I describe this as a bounded null rather than an informative one. The primary interval rules out an average 2024–26 shift larger than about 6% in either direction, under the model's assumptions, and the most conservative specifications extend that lower bound to about 8%. Read as a linear ramp, the same interval corresponds to an endpoint between −8.6% and +9.4%.

This null concerns workers of all ages, so it does not contradict [1], which concerns young workers. The design shows some upward drift before 2019, though a linear-slope test does not reject it ($p = 0.27$), and the 2021 and 2022 coefficients do not jointly differ from 2019 ($p = 0.50$). Measured from 2019, a comparison that crosses the 2020 occupation-code change, the estimate is +4.2% (95% CI −6.1% to +14.6%); restricting every year to January–September leaves it at +0.2%. Education is a more serious concern, because exposure and college share correlate at 0.65 across occupation groups. Adding college share by year, an exploratory check, moves the estimate to −0.5% and widens the interval to −8.4% to +7.4%.

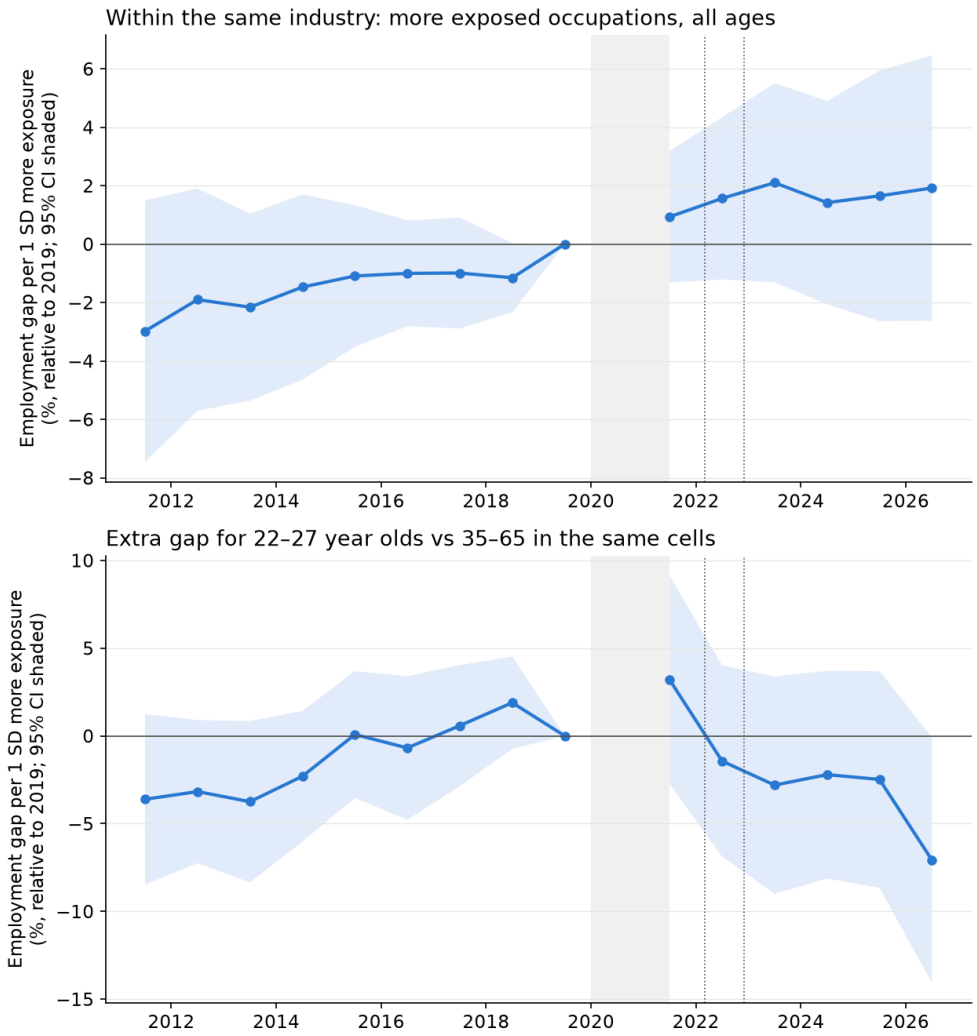


Figure 2: Test B coefficients per standard deviation of exposure, relative to 2019, with 95% confidence intervals. The top panel shows exposed versus less exposed occupations in the same sector for all ages. The bottom panel shows the additional gap for workers aged 22–27 relative to those aged 35–65. Multiplying the change between 2022 and 2024–26 by the quartile gap of 2.55 (top) or 2.53 (bottom) gives the headline estimates.

6.3 Young Workers

If AI is displacing entry-level work, the effect should appear here first. The two main sources agree on direction but not on timing.

In QWI, hires of 22–24 year olds relative to 35–44 year olds fall by 9.3% (95% CI −18.3% to −0.2%) in exposed industries compared with less exposed industries. Timing is the obstacle to attributing this to AI. The primary specification drops the pandemic quarters, but the version that retains them and the late-2021 baseline (+0.1%) both indicate that the gap opened in 2020–21. It then held nearly constant, at −10.2%, −8.6%, −8.9% and −9.3% across 2022–25. As Figure 3 shows, the series also drifts downward for years before the pandemic. A model released in late 2022 cannot explain a gap already in place, although a stable gap cannot rule out AI preventing a recovery that would otherwise have occurred.

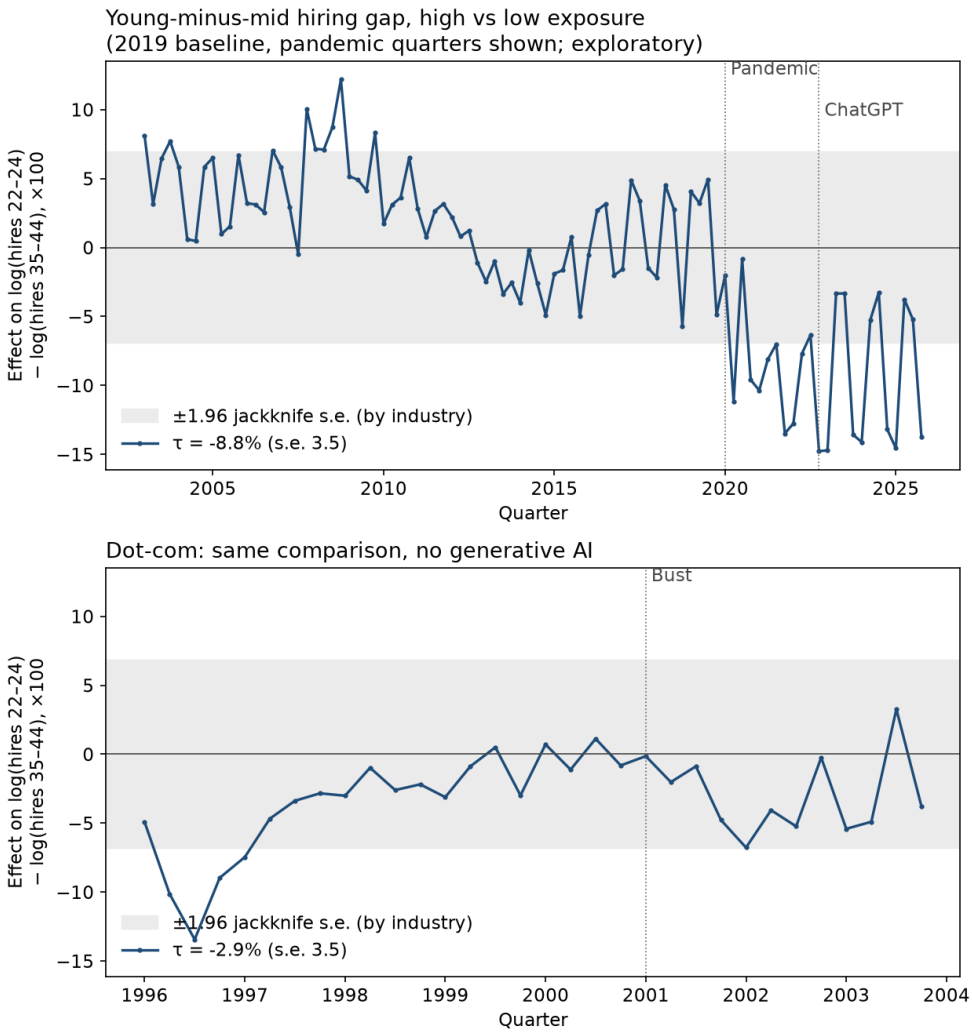


Figure 3: QWI hiring gap between young and mid-career workers, high versus low exposure. This exploratory version retains the pandemic quarters for display (estimate −8.8%); the primary specification drops them (−9.3%). The dot-com comparison is below.

In the CPS, the triple difference estimates that young workers fell 6.3% relative to older workers in the same occupation and sector (95% CI −15.5% to +2.9%, p = 0.178), with a 91% probability of a negative effect under a flat prior. The pre-trend test passes (p = 0.18), and the direction matches [1]. The yearly coefficients per standard deviation of exposure are −1.4%, −2.8%, −2.2%, −2.5% and −7.1% across 2022–26. The largest move therefore falls in 2026, a partial year of noisy survey data. Round 1 SDiD on the same survey gives a similar reading: −12.6% (95% CI −26.8% to +1.6%), largely in place by 2021, with a possible widening in 2026. Excluding 2026 reduces the triple-difference estimate to −2.3% (p = 0.64), which shows how much depends on that one year.

The young-worker estimate is sensitive to reasonable specification choices. Table 7 reports exploratory checks, none of which is pre-registered or corrected for multiple testing. The sign is consistent, but the size ranges from about −1% to −11%, and only the version that removes cell trends is significant.

Table 7: Young versus older workers under alternative specifications (exploratory).

Specification	Estimate	95% CI	p
As reported	−6.3%	−15.5% to +2.9%	0.18
Excluding 2026	−2.3%	−11.9% to +7.3%	0.64
Age-interacted cyclicality and rate sensitivity added	−4.7%	−14.1% to +4.6%	0.31
January–September in every year	−2.7%	−13.6% to +8.2%	0.63
Respondent threshold per surveyed month	−6.2%	−15.0% to +2.6%	0.16
Cell filter through 2019 only	−6.1%	−14.4% to +2.2%	0.15
Measured from 2019 rather than 2022	−9.9%	−25.3% to +5.5%	0.20
Each cell's linear trend removed	−11.4%	−21.3% to −1.4%	0.026
College share by age and year added	−5.5%	−16.2% to +5.2%	0.31
Same cells only, young minus older within each cell	−0.9%	−10.3% to +8.5%	0.85

These checks move the estimate for different reasons. Adding controls reduces it, consistent with a general hiring slowdown falling harder on young workers. Using the same months each year also reduces it. The cell filter matters little. Removing cell trends increases it, because young workers' relative position in exposed occupations appeared to improve before 2019, although the slope test does not reject a flat trend ($p = 0.27$) and extrapolating cell-level trends is noisy. The largest change comes from comparing young and older workers only within the same occupation-sector cells, which absorbs everything that hits a cell in a given year: the estimate falls to −0.9%. The reported version compares 167 young cells with 484 older cells, most of them different, so part of its gap reflects which cells are compared. Clustering by both occupation group and sector, rather than by group alone, narrows the reported interval to −13.1% to +0.5% ($p = 0.065$); for Test B it widens to −7.2% to +7.7%.

I keep −6.3% as the reported estimate because it was specified in advance, and choosing among these versions after seeing them would be the kind of forking path Section 5 warns about. The reasonable reading is that the point estimates are negative, but their size cannot be pinned down with these data.

The dot-com bust is of limited help on this question. Neither estimate shows a clear tilt against young workers (−2.9% in QWI, 95% CI −9.7% to +3.9%, using only 20 states; +0.6% in CPS, 95% CI −13.1% to +14.2%), but both intervals are too wide to rule out one of similar size.

6.4 Alternative Explanations

The timing favors the alternatives for the broad industry decline, but this study does not separate them from one another. Rate hikes, the boom's end and Section 174 all predate ChatGPT, and the industry-level pattern resembles a boom unwinding. Test B absorbs sector-wide shocks and finds no decline, though not occupation-specific ones, while the rate and cyclicality controls in Test A allow for differential sensitivity across industries. The post-2022 pattern, in other words, does not require AI to explain it. This study does not estimate how much each alternative contributes, and it does not test Section 174 directly, although excluding computer occupations and the information and professional sector in Table 6 offers a partial check.

6.5 Statistical Power

Table 8: What each design can detect.

Design	Injected effect	Result
Test A (ramp)	Ramp reaching 10% by 2026 Q3	94% power; analytic 80%-power threshold about 8%
Test B (ramp)	Ramp reaching 10% by the final period	88–90% power on a shuffled exposure score; using the actual standard error of 3.0%, the 80%-power threshold is about 13%
SDiD, industries (step)	10% step from 2023	100% power, with correct dating
SDiD, QWI and CPS by age (step)	15% step from 2023	About 40% power; detectable effect about 20–24%
Triple difference, CPS	None; no injection test was run	Standard error of 4.6% implies an 80%-power threshold of about 13% on the estimand (analytic only)

The consequence is an asymmetry. Test A clears the pre-registered power rule. Test B clears it only when power is measured on shuffled exposure, although its confidence interval still bounds the average 2024–26 shift at about 6%. The age designs are weaker. "No clear widening for young workers since ChatGPT" is therefore weak evidence in either direction, and the Round 1 step injections likely flatter these designs, since real adoption is gradual.

6.6 Verdict

The pre-registered outcome is Verdict 4, "can't tell yet." Test A fails its pre-trend test ($p = 0.073$), and under the rules a design that fails that test is reported as failed rather than interpreted, even though its estimate would otherwise clear the Holm threshold. Test B passes its checks and shows no meaningful change. Verdict 3 requires both primaries to pass, and Verdict 1 requires a Holm-significant negative estimate from a design that passes its checks. Four points matter more than the label:

- **The label is fragile.** Across six versions of the industry exposure score, Test A's pre-trend p-value ranges from 0.067 to 0.107 and only one passes. The estimate is steadier, between −3.7% and −5.0%.
- **The industry decline is hard to attribute.** About 40% of it disappears when industries with similar pandemic booms are compared, and the same model shows a −3.9% decline in the dot-com bust, when there was no generative AI.
- **Within sectors, there is no sign of a large average change,** although Test B clears the power rule only on shuffled exposure. Using its actual standard error, the 80%-power ramp is about 13%.
- **It is not evidence of no effect.** Test A's estimate is negative with an interval that excludes zero, and young workers' estimates lean negative, although they are not statistically clear and one comparison puts them near zero.

7. Verification and Reproducibility

A mistake in code or data can produce a number that is precise, plausible and wrong. Before publishing, I tried to break the analysis rather than confirm it: I checked the estimators against independent software, the data against its sources, the reported numbers against the saved output, and the whole pipeline by rerunning it from raw data. Table 9 summarizes the checks, and `docs/VERIFICATION_AUDIT.md` in the repository gives the details and the script behind each.

Table 9: Verification checks.

Check	Compared with	Result
Estimators on simulated data	Planted effects, and regressions with every fixed effect entered as a dummy	Synthetic control, SDiD and the event study recover the planted effects; the event study matches the full dummy regression
Event studies (Tests A and B, age version)	An independent refit in pyfixest	Point estimates match to four decimals; standard errors differ by 1–4% because of a small-sample convention
Wild cluster bootstrap	The wildboottest package	p-values agree within simulation error
Synthetic DiD	The synthdid package	Estimates within 0.02 percentage points; unit weights correlate at 0.999
CPS extract	The raw IPUMS file and published BLS series	Sampled rows match the raw file; employment at ages 25–54 is within 0.7% of BLS in every month since 2010
CES and QWI	FRED and the Census API	Spot checks of 10 CES series and one QWI cell match exactly
Industry exposure scores	Each industry's matched Census industries, printed and read	Two matching errors, now corrected
January population controls	Other months of the CPS	January shifts in 2024–26 are no larger than in earlier years
Reported numbers	The saved results	An automated test checks the estimates, intervals and p-values in the text and tables; all match
Full rerun	A fresh copy and software environment, starting from raw data	All 6,861 saved values reproduced exactly

The checks were worth running because they found problems, each corrected in the results above. The pyfixest refit exposed a bug that had left the age model without its sector-by-age-by-year effects; fixing it moved the estimate from −6.8% ($p = 0.087$) to −6.3% ($p = 0.18$). Printing the industry matches found the two errors that changed Test A's verdict (Section 5). An outside review, checked against the code, found that Test A's first run departed from the pre-registered occupation mix, and an audit of the plan against the code found a pre-registered robustness check that had not been run. The number checks found one typo in a table. The README lists each correction with its date.

These checks show that the numbers are computed as described and can be reproduced. They do not show that the identifying assumptions hold, which Sections 6 and 8 discuss, and they were run within this project rather than by an independent replicator. The CPS microdata cannot be redistributed, so a full replication needs the reader's own IPUMS extract. The repository records a SHA-256 checksum for every data file, so a replicator can tell whether a different result comes from different data.

8. Limitations

The claim this study can support is deliberately narrow. The main limitations are below; the full list, with what each would take to resolve, is in `docs/LIMITATIONS.md` in the repository.

- **A spare-time project.** This is an independent analysis without peer review or specialist co-authors. I used generative AI to help write code and documents, as the README describes, so that source warrants the same scrutiny as any other.
- **Exposure is potential, not use.** I observe no data on actual AI use, so the results are closer to an intent-to-treat estimate. The scores are a 2023 snapshot, and raters disagree [3]: GPT-4 and human beta ratings agree only 65.6% of the time at the task level, although by my calculation they correlate

at 0.89 across occupations and 0.91 across the occupation groups in Test B. Across models, agreement can fall to 57% and coefficients can vary 2.4-fold [9]. That error is non-classical, so I cannot assume the true effect is larger than what I report. Only one measure was run; the pre-registered alternatives, including AIOE [16], were not.

- **Everything is relative.** Low-exposure industries still average about 0.27 on the GPT-4 rating. If AI affects all work, or shifts work from exposed jobs to comparison jobs, these designs misstate or miss it.
- **Unmeasured shocks.** Test B does not absorb shocks that differ by occupation within a sector, including Section 174, rate sensitivity [10], education, return-to-office mandates, immigration policy and tariffs. The college-share check is coarse. The age design cannot separate AI from any other slowdown in hiring.
- **The boom and AI can look alike.** "The boom unwound" and "AI prevented the boom from lasting" predict the same employment path. A mechanism test on productivity is the most direct way to separate them.
- **Pre-trends.** Test A fails its pre-trend test, and its answer moves from −5.0% to −6.1% under a linear extrapolation. With a continuous dose, comparing effects across exposure levels also needs a stronger parallel-trends assumption than a simple treated-versus-untreated comparison [17]. A full analysis under [8] is not part of this study, and passing a pre-trend test does not prove parallel trends.
- **Noisy and unsettled data.** CPS cells are small, the cell filter can drop cells that shrank, occupation codes changed in 2020 [14, 15], and CES is revised annually. Together with the pandemic, the code change roughly doubled the usual movement in occupation groups' shares from 2019 to 2021, although those moves are nearly uncorrelated with exposure (0.04). The 2026 figures cover January through September and are not seasonally adjusted, so the newest numbers are the least settled.
- **Limited power and a short window.** The age designs miss smaller effects, and an effect that is still building may not yet be detectable.
- **Limited outcomes and aggregation.** I report headcounts rather than hours, wages or job quality, and 14 broad sectors can mask changes within firms.
- **Researcher choices.** The 2019 baseline, the QWI seasonal adjustment and the pandemic exclusion came after seeing some results, as did the corrections to Test A's exposure score (Section 5). Secondary tests on actual AI use, productivity, Section 174 and earnings, the specification curves and the ACS rerun were pre-registered but not run, and no results are withheld. The employment-weighted Test A was also first omitted. The pre-registration's hash was recorded in my own notes rather than independently timestamped.

9. Conclusions and Potential Areas of Future Work

9.1 Conclusions

This paper does not show whether AI caused the decline in technology hiring, because it does not measure technology hiring directly. What it examines is total employment across occupations and private industries, compared by potential AI exposure. On that question, within the same sector, the estimated average 2024–26 change for the most exposed occupations relative to the least exposed is close to zero. Across industries, exposed work slid by an estimated 5.0%, but that design fails its pre-trend check, about

40% of the slide runs through the pandemic boom, and the same model shows a similar decline in the dot-com bust. Under the rules fixed in advance, the verdict is "can't tell yet." I am not going to argue that AI has no effect on employment, because relative comparisons of potential exposure over a short window are not sufficient evidence for that claim. What I will argue is narrower: through September 2026, these data show no large average loss, timed to AI, in exposed work's relative share of employment, and they do not rule out a larger decline emerging gradually by the end of the period. The place to keep watching is young workers, where estimates lean negative but are not statistically clear, one comparison puts them near zero, and these tests are weakest.

9.2 Potential Areas of Future Work

- **More power for young workers.** The American Community Survey offers about 50 times the CPS sample and should show whether the 2026 decline is real. This would be the highest-priority extension.
- **A mechanism test.** BLS productivity data can separate the two stories in Section 8. If output holds while hours fall, that points toward labor-saving technology; if both fall, it points toward a demand shock.
- **Actual use.** Replacing potential exposure with usage data from the Anthropic Economic Index and the Census Business Trends and Outlook Survey would reduce measurement error, though usage is measured after the fact and raises reverse-causality concerns.
- **Section 174.** R&D-intensive industries should fall from 2022 and recover from 2025 if the tax change mattered, which makes the 2025 reversal a second natural experiment.
- **Specification curves.** Running 240 pre-registered combinations of exposure measure, baseline, controls, weights, pandemic treatment and sample per test would show how much the answers depend on reasonable choices.

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Appendix: Repository Guide

Material	Location
Every result on one page	`docs/SCORECARD.md`
Frozen pre-registration	`docs/PREREGISTRATION.md`

Material	Location
Design decisions and the alternatives considered	`docs/DECISIONS.md`
One notebook per analysis	`notebooks/`
Saved numbers and figures	`results/`, `figures/`
Exploratory checks in Sections 6.2 and 6.3	`scripts/test_b_filter_check.py`, `scripts/young_worker_robustness.py`
Exploratory checks on education, inference and the occupation mix	`scripts/review_robustness.py`
Corrections made to this paper and its results	`README.md`, Section 13
Repository	https://github.com/pbuszka1/ai-or-interest-rates
Full list of limitations	`docs/LIMITATIONS.md`
Status of every pre-registered item	`docs/PREREGISTRATION_STATUS.md`
Comparison with published data	`docs/DATA_VALIDATION.md`
Independent re-estimation in pyfixest	`scripts/reestimate_pyfixest.py`
Verification checks in Section 7	`docs/VERIFICATION_AUDIT.md`
Full rerun from raw data, and its comparison with the saved results	`scripts/run_all.py`, `scripts/compare_rerun.py`, `results/rerun_comparison.json`
Checksum of every data file	`docs/DATA_MANIFEST.csv`